

Borrowed Size, Agglomeration Shadows and Cultural Amenities in North-West Europe

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ABSTRACT *It has been argued that the concept of “borrowed size” is essential to understanding urban patterns and dynamics in North-West Europe. This paper conceptualizes this idea and provides an empirical exploration of it. A place borrows size when it hosts more urban functions than its own size could normally support. A borrowed size for one place means that other places face an “agglomeration shadow” because they host fewer urban functions than they would normally support. This paper explores the extent to which size and function are related for places in North-West Europe and tries to explain why one place borrows size while the other faces an agglomeration shadow by examining the position of places within the regional urban system. The presence of urban functions was approximated using high-end cultural amenities. We conclude that the largest places in their functional urban area (FUA) are better able to exploit their own mass. The largest place in a FUA is also better able to borrow size from nearby places and from (inter)national urban networks than the lower-ranked places.*

1. Introduction

1.1. Rise of the Consumer City

In the past, cities were primarily seen as centres of production, while nowadays they are increasingly perceived as centres of consumption (Glaeser *et al.*, 2001; Rappaport, 2008). Although nominal wages increase with the city size, the costs of living increase even more, indicating that people are willing to give up real wages to enjoy the consumer amenities that are typically found in large cities (Tabuchi & Yoshida, 2000). This trade-off holds

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especially for the higher-educated part of the workforce (Lee, 2010; Dalmazzo & De Blasio, 2011).¹ Amenities, here, are not only the aesthetic properties of large cities such as historical buildings and heritage sites (e.g. Van Duijn & Rouwendal, 2013), but also the presence of a variety of specialized goods and services (e.g. Berry & Waldfogel, 2010; Burger *et al.*, 2013) and the provision of specialized public services (e.g. Dahlberg *et al.*, 2012; Ebertz, 2013). Traditionally, universities, hospitals, stadiums, concert halls, theatres, galleries, museums and higher-order retail functions are present in large cities, while inhabitants of towns, villages and hamlets do not have ready local access to these amenities. The presence of consumer amenities is also increasingly important in the location decisions of firms and households, which is reflected in the higher growth rates of amenity-rich places (Glaeser *et al.*, 2001; Clark *et al.*, 2002; Chen & Rosenthal, 2008; Markusen & Schrock, 2009) as well as in the recent increase in exchange commuting in many Western societies,² that is, people living in the amenity-rich central cities and working in the suburban areas (Van der Laan, 1998; Aguilera, 2005; De Goei *et al.*, 2010; Burger *et al.*, 2011).

Consumer amenities are known to be strongly dependent on the size of the local population (Berry & Parr, 1988). In particular, the presence of a large quantity of high-end consumer functions is strongly associated with the size of a city and, as such, a manifestation of the presence of urbanization economies through consumption (Glaeser *et al.*, 2001; Rosenthal & Strange, 2004).

1.2. *From City to Urban Network*

Parallel to the rise of the consumer city, we see changes in the structure of urban systems and an increasing geographical scope for social and economic processes (see, e.g. Camagni, 1993; Batten, 1995; Champion, 2001). At the metropolitan scale, suburban areas are increasingly emerging into local centres that develop their own economic activities and, in some cases, start competing with the original urban centre. At higher spatial scales, there is an increasingly complex formation of functional linkages between historically separated urban areas. These developments are reflected in the increasing popularity of concepts such as polycentric urban regions (Kloosterman & Musterd, 2001; Parr, 2004) and mega-city regions (Hall & Pain, 2006; Florida *et al.*, 2008; Hoyler *et al.*, 2008). Such urban networks are characterized by a considerable degree of spatial integration between different places and, in many cases, complementary relationships between places (Champion, 2001; Parr, 2004; Meijers, 2007a). Accordingly, cities and towns within an urban network are thought to have different sectoral and functional specializations (Meijers, 2008; Van Oort *et al.*, 2010), where some places appear to specialize in delivering high levels of consumer amenities signified by the sometimes considerable amenity gaps between places within a region (BBSR, 2011; Silver, 2012). This study analyses how network economies allow cities to borrow size from other places at different geographical scales.³

1.3. *Borrowed Size and Agglomeration Shadows*

Within networked constellations, there is not necessarily a relationship between the size of a place and the functions it fulfils (Batten, 1995; Meijers, 2007b) because the critical mass to support particular functions can be obtained from the wider urban network. In other

words, the “impact zone” (cf. Clark, 2004) of urban functions increases.⁴ Due to the presence of spatial interdependencies, smaller places can “borrow size” and host functions that they could not have hosted in isolation. The concept of borrowed size was introduced by Alonso (1973) to explain this disconnection between the size and function of smaller cities that were part of a megalopolitan urban complex:

[t]he concept of a system of cities has many facets, but one of particular interest . . . is the concept of borrowed size, whereby a small city or metropolitan area exhibits some of the characteristics of a larger one if it is near other population concentrations. (Alonso, 1973, p. 200)

More precisely, he suggests that smaller places may “borrow” some of the agglomeration benefits of their larger neighbours, while avoiding the agglomeration costs. Even back in the 1970s, Alonso, himself hinting at the pattern of urbanization in the North-Eastern USA, noted that borrowed size was a key issue to understanding the urbanization pattern of North-West Europe: processes of borrowed size are

also quite visible . . . in certain European urban patterns, such as those of Germany and the Low Countries, whose cities, quite small by our standards, apparently achieve sufficient scale for the functioning of a modern economy by borrowing size from one another. This phenomenon transforms the issue of the size and growth of a city by redefining it to include, in some degree, its neighbours.

(Alonso, 1973, p. 200)

Despite the promise that the concept of borrowed size explains the pattern of urbanization in these countries, it has received only limited attention in economics and geography (Phelps & Ozawa, 2003).

Although smaller places can potentially host urban functions that are normally only found in larger cities through this process of borrowing size, increased spatial competition between nearby places can also lead to a situation in which a place hosts fewer functions than it could normally support. Some have conceptualized this situation as being an “agglomeration shadow” (Fujita *et al.*, 1999; Dobkins & Ioannides, 2001; Partridge *et al.*, 2009). Building on Central Place Theory (Christaller, 1933) and urban systems theory (Berry, 1964), New Economic Geography predicts a shadow effect from large cities over their surroundings, which means that the growth of areas near higher-order places will be limited due to competition effects. In terms of the presence of consumer amenities, it can be expected that places in proximity to large cities actually have fewer consumer amenities than isolated places of similar size. This expectation reflects that the provision of higher-order amenities in larger cities is often facilitated by control over the wider region as a trade area for high-end functions, where subordinate centres help to provide the minimum demand threshold to support these functions (Wensley & Stabler, 1998; Thilmann *et al.*, 2005). In this context, the overrepresentation of high-end amenities in the largest cities is driven by the advantages of locating closest to the largest customer base. Although competitors try to avoid each other, the agglomeration of amenities in large cities can be explained by savings on the travel, search and transaction costs associated with multi-purpose trips (Eaton & Lipsey, 1982; Glaeser *et al.*, 2001). Obviously, the benefits from the co-location of amenities for consumers imply

that agglomeration is to their advantage as well. Anticipating consumer behaviour, amenities that share their location with other competing and complementary amenities are more likely to attract consumers than an otherwise identical amenity located on its own. Accordingly, the agglomeration of amenities creates advantages for both the consumers and the producers of these services (Mulligan, 1984).

1.4. Research Question

In sum, increasing spatial interdependencies between places can result in a disconnection between the size of a place and its function, here measured by the presence of particular consumer amenities. On the one hand, minimum threshold demands for consumer amenities can be supported by the population of places in the urban network, which provides additional external demand. On the other hand, the presence of amenities elsewhere in the network results in external competing sources of supply drawing (potential) local consumers away from these areas (Mushinski & Weiler, 2002). However, it remains unclear which places profit and which places lose in light of increasing spatial interdependencies. In other words, which places borrow size and which places face agglomeration shadows?

In this paper, we narrow this question down to a more specific one addressing the capacity of places to borrow size. Focusing on high-end amenities in urban Europe related to cultural venues (museums, operas, theatres and public art institutions) and cultural events (music events, film festivals and art exhibitions), we examine the extent to which the population size of a place and its position in the wider regional, national and international networks affects the presence of these amenities. In particular, we are interested in how the position of a place in the regional urban network affects the relationship between the population variables and the cultural amenities present in that place. Moreover, we analyse the ability of places to borrow size by analysing national and international accessibility. The expectation is that places that are relatively well (internationally) connected are able to borrow size from other places in Europe or even beyond. So, this paper explores the question whether the position of a place in the regional urban system affects its capacity to borrow size locally, regionally and (inter)nationally.

The remainder of this paper is organized as follows. The next section presents the research approach, including concepts, data and variables, and the modelling approach. The empirical results are presented in Section 3. We conclude with a discussion of our findings in Section 4.

2. Research Approach

Cultural amenities provide a good case study for examining the occurrence of borrowed size and agglomeration shadows because their presence is generally strongly related to the size of a place. A disconnection between the size and function of a place is an indication of either borrowed size (if a place has more cultural amenities than expected given its size) or agglomeration shadows (if a place hosts fewer cultural amenities than expected given its size). We consider these concepts to be two sides of the same coin, and consequently the way to operationalize them is similar. This relationship also means that a borrowed size effect for one place implies that one or more other places are in an agglomeration shadow.

While we believe that the processes of borrowed size and agglomeration shadows can occur on many territorial scales, it would appear logical to assume that they are most often manifested on the scale of functional urban regions because nearby cities are more likely to borrow size from each other due to their proximity. At the same time, good accessibility to places beyond the own metropolitan region may also foster processes of borrowed size or agglomeration shadows on a national or international scale. For instance, the cluster of theatres in places like New York or London builds on a customer base that is perhaps even global in scope and might limit the support base for cultural amenities in the cities from which these customers come. This observation is another reason to focus on high-end cultural amenities, because consumers are willing to travel longer distances for these types of amenities. As accessibility strongly determines the ability of places to borrow size from each other, it is taken into account in this study.

2.1. Data and Variables

To analyse the moderating effect of network position on the relationship between the size of a place and the amenities that it provides, we use data from the German Federal Institute for Research on Building, Urban Affairs and Spatial Development (BBSR, 2011). This database provides information on the presence of metropolitan functions in places (mostly defined at LAU-2) in the urbanized part of Europe. In this study, we focus on the presence of cultural events and venues in 2794 West-European cities, towns and villages for the years 2006–2007. The population size of the places in the database ranges from 61 (Gerstengrund) to almost 3.5 million (Berlin). Most LAU-2 spatial units are located in France (35%) and Germany (29%), followed by Switzerland (14%), the Netherlands (8%), Austria (7%), Belgium (6%), and Luxembourg (1%).

To capture the presence of cultural amenities in a single indicator, we developed an index of cultural amenities that is based on six indicators: number of theatres, number of opera houses and music theatres, number of large music events hosted,⁵ number and quality of public art institutions,⁶ number of art fairs and film festivals, and number of art galleries. The place with the highest average score on the six dimensions receives a score of 1, while other cities receive a score relative to the highest ranked city, where the minimum score is 0. A more detailed overview of the measurement of the various indicators is provided by BBSR (2011).

Figure 1 shows the distribution of cultural amenities across European places. As can be expected, this figure shows that cultural amenities are strongly concentrated in the largest cities in North-West Europe (Berlin, Paris and Vienna; see Table 1). At the same time, most places (78%) lack high-end cultural amenities, in that they score zero on the Cultural Amenities Index, which is also signified by the low average presence of these amenities (0.0047) across all places in North-West Europe. Places with fewer than 50,000 inhabitants score, on average, 0.001 on the Cultural Amenities Index, while places with 100,000–350,000 inhabitants and places with more than 350,000 inhabitants (the 25 largest places in North-West Europe) score 0.029 and 0.257, respectively.

In this research, we examine the extent to which the presence of high-end cultural amenities in a place is first of all dependent on the local population size (LAU-2). Second, we take into account population variables that represent (1) the size of the population in the rest of the functional urban area (FUA) in which the place is situated and (2) the size of the population that can be accessed on a national and international scale (outside of the own

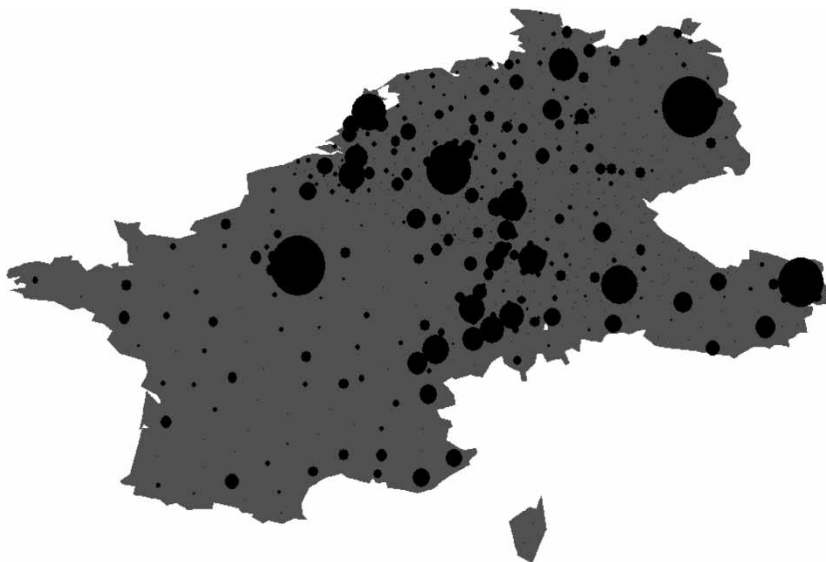


Figure 1. Geography of high-end cultural amenities in North-West Europe.

Source: Own calculations based on BBSR (2011).

Table 1. Scores of 25 largest places in North-West Europe on Cultural Amenities Index

Place (LAU-2)	Cultural Amenities Index	Population (millions)
Berlin	1.00	3.46
Paris	0.96	2.23
Vienna	0.65	1.70
Cologne	0.64	1.01
Munich	0.42	1.35
Amsterdam	0.36	0.77
Hamburg	0.29	1.79
Frankfurt am Main	0.29	0.68
Stuttgart	0.25	0.61
Bruxelles	0.23	1.09
Antwerp	0.17	0.48
Zurich	0.17	0.37
Düsseldorf	0.17	0.59
Hannover	0.12	0.52
Nürnberg	0.11	0.51
Lyon	0.10	0.48
Essen	0.09	0.57
The Hague	0.08	0.49
Marseille	0.06	0.85
Dortmund	0.06	0.58
Bremen	0.06	0.66
Rotterdam	0.06	0.59
Toulouse	0.06	0.44
Bochum	0.02	0.37
Duisburg	0.01	0.49

FUA). The question then is whether or not a place profits from being located in a larger metropolitan area or having access to a large population (inter)nationally.

Here, our definition of FUAs follows that of the ESPON 1.4.3 project (Peeters, 2011), where FUAs represent labour market regions that encompass cities, towns and villages that are economically and socially highly interdependent. The types of FUAs range from mega-city regions, such as Paris (11.1 million habitants), Milano (6.0 million inhabitants), and Dortmund (5.3 million inhabitants), to self-contained villages in more rural areas with just over 10,000 inhabitants. The average size of a place in our database is just over 30,000 inhabitants, while the average size of a place that is the largest in their FUA is almost 130,000. Overall, the places in our database are nested in 417 FUAs, meaning that each FUA consists, on average, of 6–7 LAU-2 units. By analysing the places within FUAs, local proximity is taken into account because these FUAs represent labour market regions with a maximum radius of 60 kilometres.

To capture national and international accessibility, we make use of the absolute potential accessibility to population by road, rail and air (multimodal) from the ESPON research reported in Spiekermann and Wegener (2006). This accessibility measure takes into account the population outside of the FUA weighted by the travel time to go there.⁷ The most accessible regions can be found in the mega-regions with major international airports such as the Rhine-Main (Frankfurt am Main) and Randstad (Amsterdam); the least accessible regions can be found in peripheral France, Germany and Austria.

As additional control variables, we introduce a dummy variable indicating whether a place contains a UNICEF heritage site (as an indicator of the historical presence of amenities), GDP per capita (measured at the FUA level), and country dummies. These country dummies capture, among others, institutional differences between countries with regard to the production of public goods and services, such as cultural venues. Descriptive statistics are presented in Table 2; all variables are measured for the year 2006, unless indicated otherwise.

Where a positive effect of the population on the rest of the FUA and national and international accessibility on the presence of high-end cultural amenities would indicate the presence of borrowed size effects and urban network economies, a negative effect would indicate the presence of agglomeration shadows or, in other words, spatial competition effects. Yet, we do not expect these variables to be significant. After all, one place may perhaps borrow size from its wider region or (inter)nationally, but another place may

Table 2. Descriptive statistics of variables included in the analysis ($N = 2794$)

	Mean	Standard deviation	Minimum	Maximum
Cultural Amenities Index	0.0047	0.037	0	1
Local population (100 k)	0.32	1.13	0.00061	34.61
Population rest FUA (mln)	2.16	3.07	0	11.17
(Inter)national accessibility (10 mln)	6.85	1.51	2.96	10.01
GDP per capita FUA (k)	32.39	8.47	13.00	54.00
Dummy heritage site	0.02	0.15	0	1
Dummy not largest place in FUA	0.85	0.35	0	1
• Dummy second largest place in FUA	0.11	0.32	0	1
• Dummy third largest place in FUA	0.08	0.29	0	1
• Dummy fourth largest place in FUA	0.07	0.25	0	1
• Dummy other not largest places in FUA	0.58	0.49	0	1

experience an agglomeration shadow effect. We assumed that whether a place borrows size or faces such agglomeration shadows depends on its position in the wider urban network (or metropolitan urban system). Therefore, we examine the extent to which the rank of a place within the FUA (affects and) moderates the relationship between the population variables and the Cultural Amenities Index. More specifically, we test the extent to which local population size, the population in the rest of the FUA and national and international accessibility can successfully be exploited by including an interaction term between the population variables and the rank of a place in the FUA in which it is located. Here, we assume that second or lower rank cities have more difficulties borrowing size compared to the largest city in an FUA. This assumption builds on the observation that co-location (or agglomeration) is beneficial for both retailers and consumers.

2.2. Empirical Model

Our dependent variable, the Cultural Amenities Index, can be perceived as a bounded variable that takes a minimum value of 0 (no amenities) and a maximum value of 1 (city with the highest score on the Amenities Index). Although indices (which can likewise be perceived as proportions) are often treated as if they were continuous, an estimation by ordinary least squares in a linear regression framework is likely to result in inefficient and biased estimates for the parameters. The most common way to address proportions data is the fractional logit model (Papke & Woolridge, 1996, 2008). However, this model often suffers from overdispersion if the response variable is non-binomially distributed in that the probability masses are concentrated at the boundaries. In the case of our data, many places do not have any high-end cultural amenities (in that they score zero), while only a few places score very high (see Figure 2). To account for this characteristic, we make use of a beta regression model (Kieschnick & McCullough, 2003; Ferrari & Cribari-Neto, 2004). This type of model utilizes the beta distribution, which makes it appropriate for modelling both binomially and non-binomially distributed response variables. At the same time, the interpretation of the results is similar to that for a logistic regression (Schmid *et al.*, 2013). In addition, beta regression models provide a natural way to address overdispersion by including an additional parameter (ϕ) to adjust the conditional variance of the proportion outcome. Because our proportions data contains zeros (no high-end amenities) and ones (highest ranked place), we make use of inflated beta regression models. Zero-or-one inflated beta regression involves three parts: a logistic regression model for whether or not the proportion equals 0, a logistic regression model for whether or not the proportion equals 1, and a beta regression model for the proportions between 0 and 1. Because our data particularly suffers from probability masses concentrated at 0, we predominantly focus on the zero-inflated part, herewith also acknowledging that there may be population thresholds for providing certain amenities. A more elaborate discussion of these models can be found in Cook *et al.* (2008) and Ospina and Ferrari (2012).

3. Results

3.1. Empirical Results

Table 3 reports the coefficients of the inflated beta regressions on our Cultural Amenities Index. Because we are not only interested in the statistical significance of effects but also

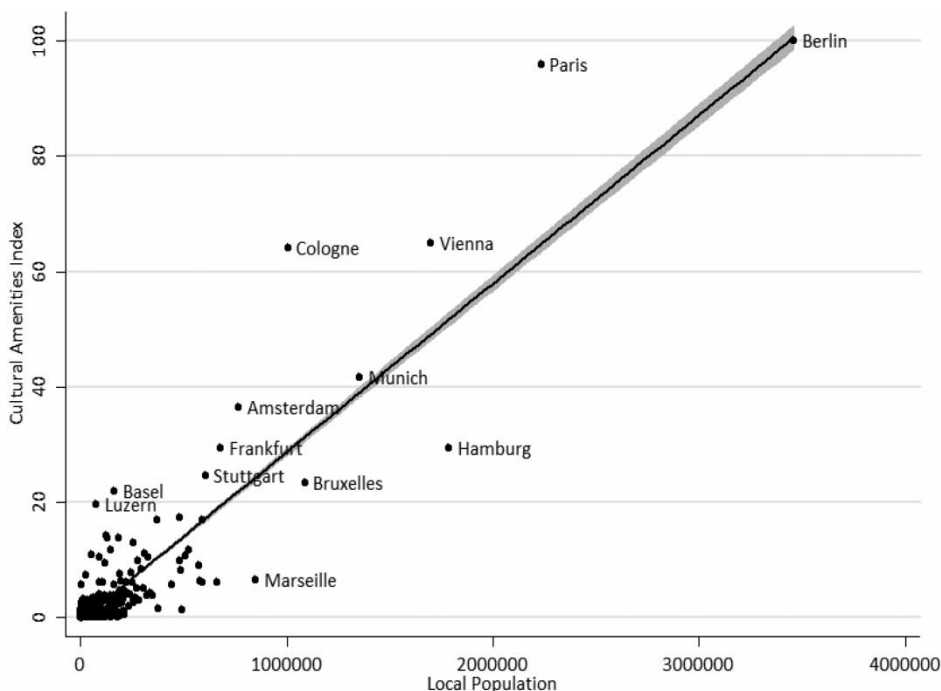


Figure 2. Relationship between local population size and cultural amenities.

in the substantive and practical significance, we also report the average marginal effects (AMEs). With regard to our population variable, the reported AMEs refer to a change in the high-end Cultural Amenities Index as a result of an assumed increase in the local population by 100,000, a change in the population living in the rest of the FUA by 1 million, and a change in (inter)national accessibility by 10 million people. Likewise, the AME for GDP per capita refers to a change in the high-end Cultural Amenities Index as a result of an assumed increase in GDP per capita by 1000 euro, while for the dummy variables, the AME refers to a change in the high-end cultural amenities variable as a result of a change in the dummy variable from 0 to 1.

Focusing on our baseline model (Table 3, Column 1), we find no effect from GDP per capita,⁸ but a positive effect from the cultural heritage site dummy on the presence of high-end cultural amenities in a place. Turning to our primary variables of interest, the results indicate that local population size is the most important determinant for the presence of high-end amenities in a place, while the population in the rest of the FUA and (inter)-national accessibility is found to be less important, which is consistent with our expectations. The AME of the local population size is 0.007, meaning that an increase in population size by 100,000 increases the high-end amenities (an index with minimum 0 and maximum 1), on average, by 0.007.

The AME is low, which signifies that there is a considerable population threshold that must be met to host high-end cultural amenities in the first place. Figure 3 shows how the AME varies across the size of the place for places between 10,000 and 1,000,000 (all observations that fall within three standard deviations of the mean, excluding outliers).

Table 3. Zero-and-one inflated beta regression on Cultural Amenities Index—regression coefficients and AMEs.

	Model 1		Model 2		Model 3	
	Coefficients	AME	Coefficients	AME	Coefficients	AME
<i>Proportion part</i>						
Local population (100 k)	0.297 (.045)**	0.0071 (.0006)**	0.248 (.036)**	0.0057 (.0005)**	0.253 (.036)**	0.0056 (.0005)**
Population rest FUA (mln)	−0.060 (.021)**	−0.0002 (.0001)	−0.018 (.021)	−0.0000 (.0001)	−0.000 (.022)	0.0001 (.0001)
(Inter)national accessibility (10 mln)	0.044 (.021)	−0.0001 (.0001)	0.079 (.030)**	0.0001 (.0001)	0.084 (.031)	0.0001 (.0001)
GDP per capita FUA (k)	0.008 (.006)	0.0000 (.0000)	0.019 (.008)**	0.0001 (.0000)*	0.022 (.008)**	0.0001 (.0000)*
Dummy heritage site	0.769 (.203)**	0.0037 (.0009)**	0.597 (.190)**	0.0029 (.0009)**	0.603 (.189)**	0.0029 (.0008)**
Dummy not largest location in FUA			−0.748 (.085)**	−0.0030 (.0004)**		
• Second largest location in FUA					−0.605 (.089)**	−0.0022 (.0004)**
• Third largest location in FUA					−0.739 (.108)**	−0.0031 (.0005)**
• Fourth largest location in FUA					−0.742 (.109)**	−0.0032 (.0005)**
• Other not largest locations in FUA					−0.927 (.110)**	−0.0040 (.0005)**
<i>Zero-inflated part</i>						
Local population (100 k)	−4.781 (.291)**		−4.579 (.352)**		−4.258 (.358)**	
Population rest FUA (mln)	−0.032 (.045)		−0.039 (.045)		−0.06 (.045)	
(Inter)national accessibility (10 mln)	0.226 (.059)**		0.214 (.061)**		0.175 (.063)**	
GDP per capita FUA (k)	0.010 (.012)		0.008 (.012)		0.002 (.012)	
Dummy heritage site	−0.702 (.398)		−0.658 (.399)		−0.693 (.401)	
Dummy not largest location in FUA			0.227 (.207)			
• Second largest location in FUA					−0.048 (.224)	
• Third largest location in FUA					0.341 (.243)	
• Fourth largest location in FUA					0.520 (.283)	
• Other not largest locations in FUA					0.574 (.243)*	
Number of Observations	2794		2794		2794	
Wald chi-square	246.0**		521.0**		548.3**	
ln phi	3.86 (.129)**		4.06 (.135)**		4.08 (.134)**	
Country dummies +	Yes		Yes		Yes	

Notes: Robust standard errors in parentheses. All models are estimated with intercept. The one-inflated part is not displayed as it is an intercept-only model. For the dummy variables the AME refers to a change of the variable from 0 to 1. AME, average marginal effect.

* $p < .05$.

** $p < .01$.

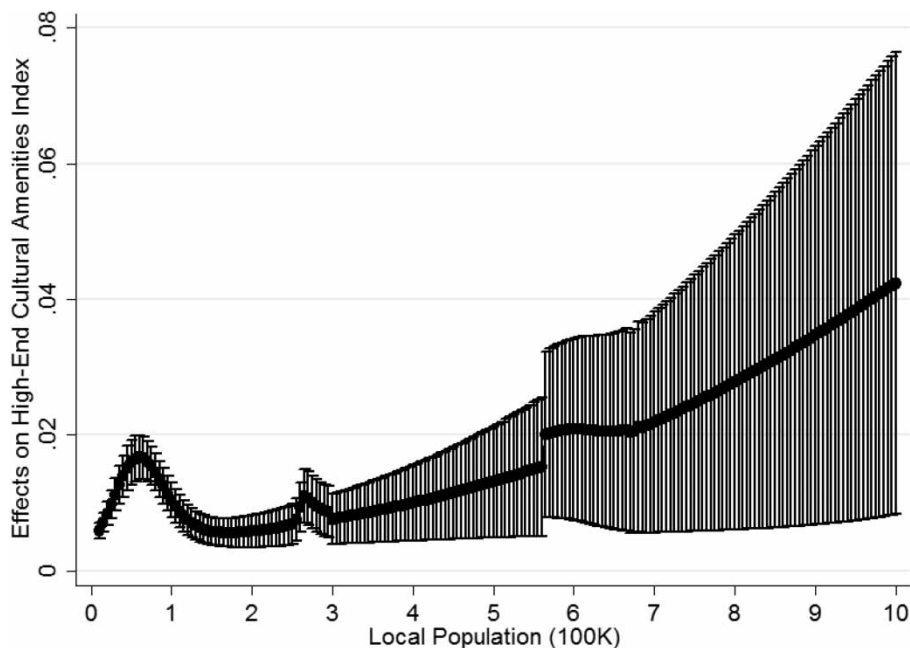


Figure 3. Fitted equation and 95% confidence interval for cultural amenities and local population size (AME).

Although an increase in population size has only a small effect on the presence of cultural amenities in places with fewer than 100,000–200,000 inhabitants, the AME of local population size on the proportion of amenities steadily increases after this threshold is reached. To exemplify, we find that for a place of 150,000–200,000 inhabitants, a 100,000 increase in the local population size would increase the Cultural Amenities Index by approximately 0.006. On the contrary, for a city of 500,000 inhabitants, 750,000 inhabitants, and 1,000,000 inhabitants, a similar increase would result in an increase in the Cultural Amenities Index by 0.013, 0.025 and 0.042, respectively.⁹ A spike in the graph can also be observed for places having between 10,000 and 100,000 inhabitants, indicating that the high-end Cultural Amenities Index includes some amenities for which the population threshold is limited. This result is consistent with earlier work by Camagni *et al.* (1986, 2014), who indicate that when the population in a place increases and moves towards the maximum size compatible with its rank, this place becomes a potentially suitable location for higher-order urban functions, a result of achieving a critical demand size for these functions.

Consistent with our expectations, the AME from the size of the rest of the FUA and (inter)national accessibility on the score for the high-end Cultural Amenities Index is insignificant in our baseline specification. However, the position of a place in the FUA also has an effect on the presence of high-end cultural amenities (see Models 2 and 3 in Table 3). Not being the largest city decreases the cultural amenities by approximately 0.003 on average, holding all other variables constant. Likewise, the lower the rank of a place in its FUA, the fewer high-end amenities are present in that place. Compared to

being the largest city in the FUA and holding everything else constant, the second largest places score, on average, 0.002 lower on the Cultural Amenities Index, while the fifth or lower-ranked places score, on average, over 0.004 lower. Accordingly, it can be inferred that the largest places within a FUA fulfil a central place function with regard to hosting high-end cultural amenities and, hence, borrow size from elsewhere, thereby casting an agglomeration shadow over the surrounding places.

At the same time, the rank of a place in a FUA may also affect the extent to which it can exploit its local size as well as the extent to which it can profit from the size of the rest of the FUA and from national and international accessibility. To test whether this situation is indeed the case, we estimated additional regressions, including interaction terms between the population variables and the dummy variable indicating whether a place is the largest within an FUA.¹⁰ Because interaction effects are generally difficult to interpret in non-linear models, we graphically show how the effect of the position in the FUA on the presence of high-end cultural amenities depends on local population size, population in the rest of the FUA, and national and international accessibility. The regression table can be found in Appendix 1.

Figures 4–6 graphically show how the position in the FUA affects the capacity to exploit local potential (Figure 4) or the potential to borrow size (exploit) from the critical mass present in the wider FUA (Figure 5) or (inter)nationally (Figure 6). Several conclusions can be drawn from these graphs. First, places are better able to profit from being the largest in their FUA when they are small (Figure 4). Estimating the AMEs for being the largest place in the FUA for different place sizes,¹¹ we find that the AME of local population size on the Cultural Amenities Index is stronger for places that are the largest in their FUA. At the same time, we see that the positive moderating effect of

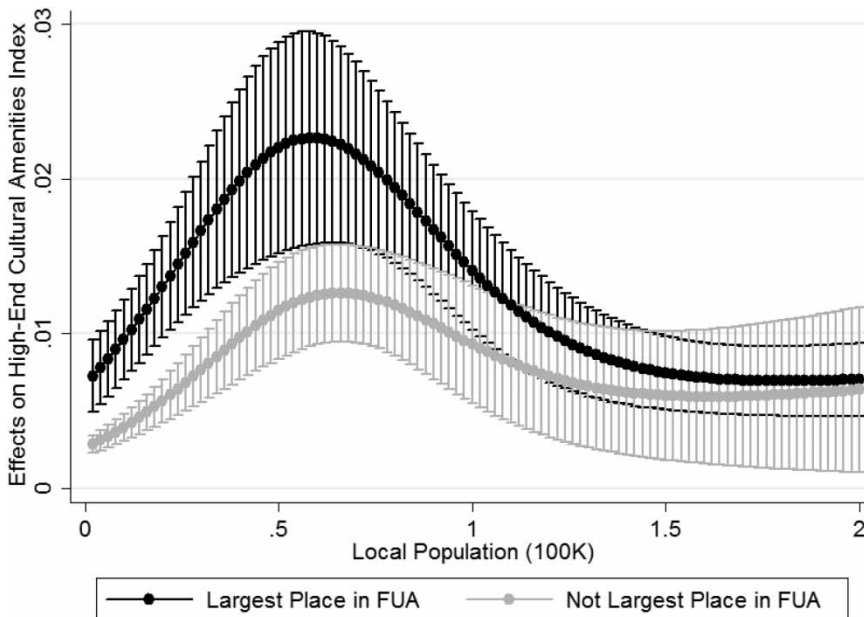


Figure 4. Exploiting local potential—AMEs.

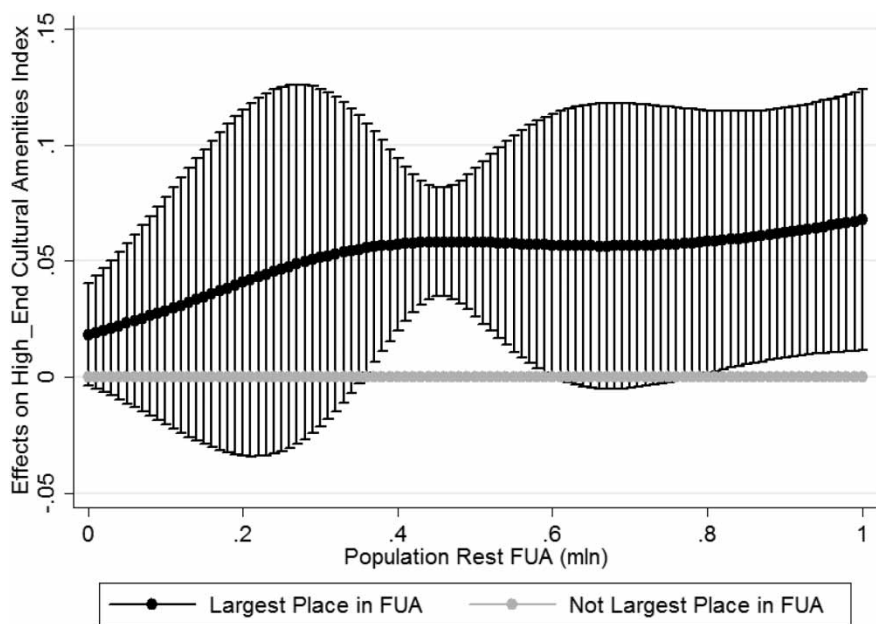


Figure 5. Exploiting regional potential—AMEs.

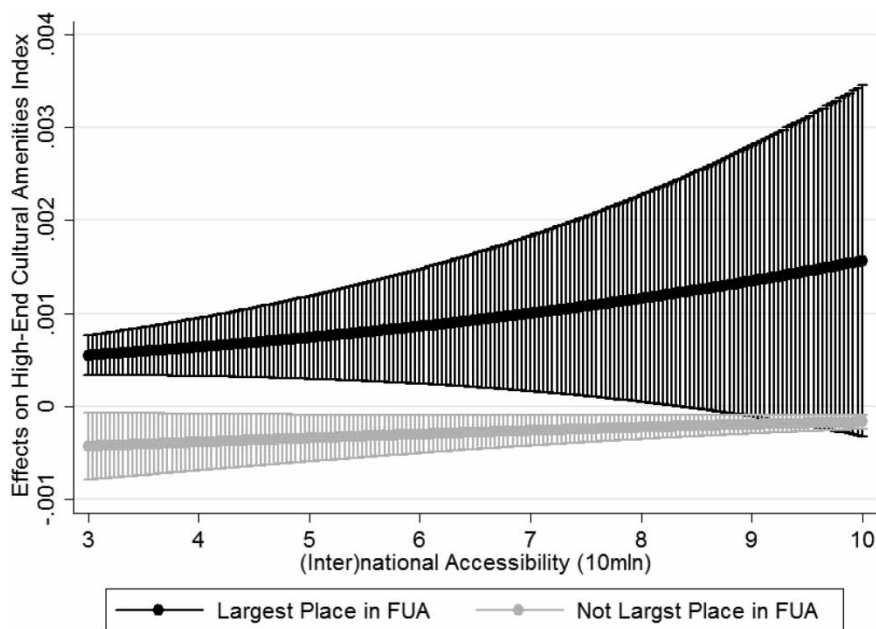


Figure 6. Exploiting national and international accessibility—AMEs.

being the largest place in an FUA on the relationship between local population size and the presence of high-end cultural amenities is the most pronounced for places with between 40,000 and 80,000 inhabitants.

Second, places tend to profit more from the size of their FUA when they are the largest in their FUA (Figure 5).¹² For the places that are the largest in their FUA, a 1 million increase in population in the rest of the FUA would result, on average, in an increase of 0.05 in the high-end Cultural Amenities Index. On the contrary, the size of the rest of the FUA does not affect the presence of high-end cultural amenities in other places in the FUA.

Third, (inter)national accessibility has a much more positive impact on the presence of high-end cultural amenities in the largest places in the FUAs (Figure 6). From this result, it can be concluded that the largest places in an FUA profit most from (inter)national accessibility and borrow size in (inter)national urban networks to support the production of high-end cultural amenities. At the same time, it should be noted that the significance of this finding is limited: an increase in (inter)national accessibility by 10 million would, even for the largest places in the FUAs, only result in an increase in the Cultural Amenities Index by 0.001–0.002. To exemplify, the most accessible places in North-West Europe would score, on average, 0.007–0.014 higher on the Cultural Amenities Index than the least accessible places, holding everything else constant.

4. Conclusion

It has been argued that the concept of “borrowed size” is essential for understanding urban patterns and dynamics in North-West Europe. This paper is one of the first to empirically explore the concept of “borrowed size”. It is likely that a place borrows size when it hosts more urban functions than its own size could normally support. The opposite, however, also occurs: a place can host fewer urban functions than it should normally be capable of supporting. Often, this situation is due to competition from nearby places or from places with good (inter)national accessibility. To capture this phenomenon, we proposed using the concept of “agglomeration shadows” and presented it as the antonym of “borrowed size”.

In this study, the presence of urban functions was measured using information on high-end cultural amenities such as theatres, operas, public art institutions, concerts, and art and film festivals. We examined the extent to which local population size, on the one hand, and population potential elsewhere in the FUA or even beyond, on the other hand, affects the presence of these amenities. The empirical results indicate that, as expected, the size of a place is the most important factor explaining the presence of cultural amenities. Whether a place is located in a small or large FUA or has high (inter)national accessibility, giving access to a large (international) population potential, mattered less. This result indicates that the geographical scope of consumption is still very local. However, the presence of high-end cultural amenities in a place is also dependent on the position of that place in its FUA. To capture this position, we used measures that refer to the rank of a city within its functional urban region. The results show that cultural amenities are disproportionately concentrated in the places that are the largest in their FUA and that places profit more from being the largest in their FUA when they are small in terms of population size. In addition, places profit more from the size of the rest of the FUA and (inter)national accessibility when they are the largest place within their FUA.

In sum, the distribution of high-end cultural amenities in North-West Europe still follows a Christallerian logic, signified by the concentration of these consumer services within the largest place in a FUA as a result of the higher level of local demand and the accessibility of these often central places to clients from neighbouring places. At the same time, the places that lie in the shadow of these central places are confined to lower levels of high-end cultural amenities than could be expected given their size. In other words, on average, larger cities cast a shadow over smaller neighbouring cities (as predicted by the New Economic Geography) rather than these smaller cities borrowing size from their larger neighbour (as suggested by Alonso). This study indicates that the largest cities in a FUA profit more from the size of the rest of the area in which they are located and from having access to a large (inter)national population. The lower order places, which are not the largest city in the FUA, are less likely to borrow size from other places because they are more likely to lie in the shadow of a first-rank city and face spatial competition effects. Accordingly, large cities profit most from borrowed size effects, whereas small- and medium-sized places in Europe typically flourish in isolation.

The strong presence of agglomeration economies in consumption questions the extent to which small- and medium-sized cities can compete with large cities on the basis of cultural amenities. All the same, a caveat of this research is that we have predominantly examined high-end cultural amenities; small- and medium-sized cities near large urban agglomerations do not necessarily have to be confined to lower-end cultural amenities because these economic activities can follow a different location pattern. Although Madonna, Radiohead, Lady Gaga, U2, and Justin Bieber might avoid small- and medium-sized cities, more locally and regionally oriented musicians do not. Likewise, small- and medium-sized cities typically lack film and art festivals, but can host smaller-scale cultural events. Hence, future research should also try to incorporate other types of consumer services. We can extend this observation even further, as future research could focus on a much broader variety of urban functions beyond only high-end cultural amenities. It could well be that the complementarity of urban functions between places comprising a metropolitan region is the key to understanding processes of borrowed size and agglomeration shadows. Perhaps a place faces an agglomeration shadow in one respect, but borrows size in another. Likewise, future research could focus on using more detailed measures for the position of places in areas and networks, also examining the extent to which being the largest place in the wider region or country affects the presence of urban functions. Moreover, it is interesting to analyse the multiple factors that determine the ability of some places to derive critical mass from other regions. Here, one can not only consider the quality of infrastructure and ease of access (see also Agnoletti *et al.*, 2014) but also other factors, such as the general quality of the services provided and the overall level of demand and supply.

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Notes

1. This viewpoint typically builds on the Rosen (1979) and Roback (1982) approach.
2. For a criticism of the amenity-growth nexus, see Peck (2005) and Storper and Scott (2009).
3. See also the study by Giffinger and Suitner (2014), which distinguishes between three different levels of integration of urban agglomeration, namely the micro, meso and macro levels. Micro-level polycentricity points at the level of internal integration of metropolitan regions, meso-level polycentricity defines dense national urban networks and macro-level polycentricity describes the embedding of metropolitan regions in wider transnational or global networks of metropolises.
4. This increased impact zone is consistent with observations that small- and medium-sized places have joined the urban competition for investments and labour (Smidt-Jensen *et al.*, 2009).
5. This indicator is based on the tour dates for bands and orchestras covering a broad range of musical styles: The Rolling Stones, Madonna, Sting and Bon Jovi (pop and rock); Anna Netrebko, Anne-Sophie Mutter, the Vienna Philharmonic and the New York Philharmonic Orchestra (classical music); and Cats (musical) as well as annual international jazz festivals.
6. Public art institutions include public art museums and galleries, non-profit or private art clubs, art foundations and associations, art archives and related libraries, and art colleges or universities teaching art, cinematic art, photography, design and graphic arts.
7. Although the multimodal accessibility measure was originally developed for NUTS-3 regions, it has been adapted to FUAs for this research.
8. Here, it must be mentioned that our measure of GDP per capita is calculated at the FUA level, whereas the Cultural Amenities Index is calculated at the place level, meaning that our GDP variable does not capture the, in some cases, strong heterogeneity in income between nearby places (e.g. centres and sub-urban places). However, income that is generated in one place can easily be spent in another place, and a measure of income at the FUA level is better able to control for this. When we control for the position of a place in a FUA, we find a small positive AME of GDP per capita, where an increase of GDP per capita by 1000 euro in the FUA results in an increase in the high-end Cultural Amenities Index by 0.0001.
9. The results of these estimations are available upon request.
10. Similar regressions were run for interactions between the rank order and the local population, the population in the rest of the FUA and national and international accessibility. For brevity, these graphs are not displayed. However, these estimations do not lead to different conclusions regarding borrowed size and spatial competition effects.
11. Because the size of the largest place that is not the largest in its FUA (Oberhausen, Germany) is just over 200,000, we do not show the AMEs for larger local population sizes in this graph.
12. Here, we focus on regional hinterlands with up to 1 million inhabitants because there are few places (less than 0.5%) that are the largest in their FUA for which the size of the rest of the FUA is more than 1 million inhabitants.

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Appendix 1. Zero-and-one inflated beta regression on Cultural Amenities Index—interaction effects

	Coefficients		
	Model 4 (Graph 4)	Model 5 (Graph 5)	Model 6 (Graph 6)
<i>Proportion part</i>			
Local population (100 k)	0.389 (.103)**	0.202 (.044)**	0.247 (.034)**
Population rest FUA (mln)	−0.020 (.021)	−0.031 (.025)	−0.015 (.021)
(Inter)national accessibility (10 mln)	0.073 (.031)*	0.059 (.030)*	−0.015 (.026)
GDP per capita FUA (k)	0.020 (.008)**	0.019 (.008)*	0.023 (.008)**
Dummy heritage site	0.594 (.190)**	0.588 (.186)**	0.630 (.190)**
Dummy not largest location in FUA	0.787 (.093)**	0.692 (.087)**	−0.441 (.317)
Dummy not largest location in FUA*Local population (100 k)	−0.131 (.110)		
Dummy not largest location in FUA*Population rest FUA (mln)		1.298 (.251)**	0.176 (.046)**
Dummy not largest location in FUA*(Inter)national accessibility (10 mln)			
<i>Zero-inflated part</i>			
Local population (100 k)	−4.534 (.408)**	−4.491 (.364)**	−4.562 (.342)**
Population rest FUA (mln)	−0.038 (.045)	−0.039 (.044)	−0.044 (.044)
(Inter)national accessibility (10 mln)	0.214 (.061)**	0.217 (.061)**	0.259 (.063)**
GDP per capita FUA (k)	0.008 (.012)	0.008 (.012)	0.008 (.012)
Dummy heritage site	−0.659 (.397)	−0.665 (.398)	−0.680 (.408)
Dummy not largest location in FUA	−0.299 (.330)	−0.109 (.226)	1.427 (.712)*
Dummy not largest location in FUA*Local population (100 k)	0.187 (.681)		
Dummy not largest location in FUA*Population rest FUA (mln)		−7.251 (6.56)	−0.279 (.118)*
Dummy not largest location in FUA*(Inter)national accessibility (10 mln)			
Number of observations	2794	2794	2794
Wald chi-square	538.4**	665.4**	600.2**
ln phi	4.06 (.135)**	4.19 (.122)**	4.11 (.135)**
Country dummies +	Yes	Yes	Yes

Notes: Robust standard errors in parentheses. All models are estimated with intercept. The one-inflated part is not displayed as it is an intercept-only model. Since average marginal effects for the interaction effects are not provided, these are not displayed.

* $p < .05$.

** $p < .01$.

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